

The endpoint Orlicz loss persists for every exchangeable selector

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Abstract

Burstein proves an upper bound with loss $(\log \log n)^{\alpha/2}$ when selecting $n/(e \log^{\alpha+1} n)$ functions from a bounded orthogonal system, and shows that independent Bernoulli selection cannot improve this loss. We extend the obstruction from Bernoulli thinning to every exchangeable (permutation-invariant) random selector that returns at least the target cardinality. Along an explicit sequence of dimensions, the trigonometric system incurs the full loss with probability tending to one. The proof uses sampling without replacement, negative correlation of disjoint full-block events, a second-moment inequality, and a Dirichlet-kernel lower bound in the Luxemburg norm. This is a substantial randomized-selection subcase; it does not resolve the unrestricted deterministic existence question.

1 Source question

The source is Will Burstein, Λ_p *Style Bounds in Orlicz Spaces Close to L^2* , arXiv:2505.00155v2 [1]. On page 13, after Theorem 2.10 and in Remark 2.11, the paper states that the optimality of its equation (2.4) remains open. The relevant source passage is reproduced below.

$$\|f\|_{\Phi}/\|a\|_2 \leq K(\alpha) \cdot \log^2(u_0 + S^* \log n),$$

which completes the proof. $\square$

2.3. Probabilistic Sharpness Result. The result of this section uses some ideas similar to those of the probabilistic sharpness result of [2]. As before, constants can vary line by line.

Theorem 2.10. *Fix $\rho > 0$ and $\alpha > 0$ and set $n := \frac{m}{2\rho} m^m$. Consider the probability space $[0, 1]$ with Lebesgue measure. For $k \in [n]$, we set*

$$\varphi_k := \exp(-2\pi i k(\cdot)),$$

the orthonormal characters on $[0, 1]$. Let $\{\zeta_k\}_{k=1}^n$ be $\{0, 1\}$ -valued and i.i.d. with mean $\delta = \frac{1}{\log^{\rho}(n)}$. Let J_{ω} be the random set,

$$J_{\omega} := \{k \in [n] : \zeta_k(\omega) = 1\}.$$

Let $\Phi(u) = u^2 \log^{\alpha}(u)$ for $u \geq u_0$, and $\Phi(u) = c(\alpha)u^2$ otherwise. $u_0 \geq e$ and $c(\alpha)$ are constants, depending on α , chosen so that Φ is a Young function. Then,

$$(2.13) \quad \mathbb{E}_J \left(\sup_{\|a\|_2 \leq 1} \left\| \sum_{k \in J} a_k \varphi_k \right\|_{\Phi} \right) \geq K(\alpha, \rho) \log^{\frac{\alpha}{2}}(\log n).$$

Remark 2.11. We state our main result, Theorem 2.7, in terms of a probability over subsets of $[n]$ following a Bernoulli distribution. We could have, just as easily, stated our main result in terms of an expectation over subsets of $[n]$ as in equation 2.13. The latter form, using the expectation, is the formulation of Theorem 1.5 of [12] (we note that the main technical machinery of Theorem 1.5 is Lemma 4.3, of [12], which uses similar techniques to Theorem 2.7). Hence, we can rule out the common randomized technique, subset selection of $[n]$ by a Bernoulli distribution on $[n]$'s indices, to show the existence of a subset which sharpens the bound of equation 2.4. The question of whether the bound of equation 2.4 is optimal remains open, but equation 2.13 gives evidence that the bound is indeed optimal.

Fix $\alpha > 0$. Let Φ be the Young function used in the source: for fixed constants $u_0 \geq e$ and $c(\alpha) > 0$,

$$\Phi(u) = u^2 \log^{\alpha} u \quad (u \geq u_0), \quad \Phi(u) = c(\alpha)u^2 \quad (u < u_0).$$

For $J \subseteq [n]$, define the Fourier synthesis norm

$$\mathcal{N}_{\Phi}(J) = \sup_{\|a\|_2 \leq 1} \left\| \sum_{j \in J} a_j e^{2\pi i j x} \right\|_{L^{\Phi}(\mathbb{T})},$$

where the Orlicz space has its Luxemburg norm. The source proves a matching lower bound for an independent Bernoulli set. The open question asks whether the logarithmic loss is necessary for the existence of a subset, without restricting how that subset is chosen.

2 Proof intuition

The Bernoulli lower bound in the source partitions the frequency interval into short consecutive blocks. If a random set contains one complete block, then the normalized Dirichlet kernel on that block has Orlicz norm of order $(\log \log n)^{\alpha/2}$. Independence makes a full block likely.

Independence is not essential. If one samples a fixed number of indices without replacement, disjoint full-block events are negatively correlated. Their expected number tends to infinity on a suitable dimension subsequence, and the elementary second-moment inequality then makes a full block occur with probability tending to one. An exchangeable selector is uniform after conditioning on its cardinality. Monotonicity therefore transfers the same obstruction to every exchangeable selector with at least the target size.

What remains open is precisely the non-exchangeable, label-dependent choice: a specially constructed deterministic subset may avoid every full consecutive block, so this proof does not establish deterministic optimality.

3 The partial theorem

Recall that a random subset $J \subseteq [n]$ is *exchangeable* if its law is invariant under every permutation of $[n]$.

Theorem 1. *Fix $\alpha > 0$ and put $\rho = \alpha + 1$ and $C = 2(\rho + 1)$. For integers $m \geq 2$, set*

$$q_m = \lceil Cm \rceil, \quad B_m = m^{q_m}, \quad n_m = mB_m,$$

and

$$k_m = \left\lceil \frac{n_m}{e(\log n_m)^\rho} \right\rceil.$$

For each m , let $J_m \subseteq [n_m]$ have an arbitrary exchangeable law satisfying $|J_m| \geq k_m$ almost surely. Then there is a constant $c_\alpha > 0$ such that

$$\mathbb{P} \left\{ \mathcal{N}_\Phi(J_m) \geq c_\alpha (\log \log n_m)^{\alpha/2} \right\} \longrightarrow 1 \quad (m \rightarrow \infty).$$

In particular, no exchangeable random selection rule at the source cardinality can replace the factor $(\log \log n)^{\alpha/2}$ by $o((\log \log n)^{\alpha/2})$, even along all sufficiently large members of this explicit dimension subsequence.

We separate the analytic and probabilistic ingredients.

Lemma 2 (A complete consecutive block is bad). *There are $c_\alpha > 0$ and $m_0(\alpha)$ such that, whenever $m \geq m_0(\alpha)$ and J contains m consecutive integers,*

$$\mathcal{N}_\Phi(J) \geq c_\alpha (\log m)^{\alpha/2}.$$

Proof. Let the block be $\{r + 1, \dots, r + m\}$ and take coefficients $m^{-1/2}$ on the block and zero elsewhere. Multiplication by $e^{2\pi i r x}$ does not change the absolute value. For $0 \leq x \leq 1/(8m)$, every real part in $\sum_{j=1}^m e^{2\pi i j x}$ is at least $1/\sqrt{2}$. Hence the normalized sum f satisfies

$$|f(x)| \geq \sqrt{m/2} \quad \text{on a set of measure } \frac{1}{8m}.$$

Set $w = c(\log m)^{\alpha/2}$, where $c > 0$ will be fixed sufficiently small. For large m , $\sqrt{m/2}/w \geq u_0$ and

$$\log \left(\frac{\sqrt{m/2}}{w} \right) \geq \frac{1}{3} \log m.$$

Therefore the Luxemburg modular at scale w is bounded below by

$$\begin{aligned} \int_{\mathbb{T}} \Phi(|f|/w), dx &\geq \frac{1}{8m} \frac{m}{2w^2} \log^\alpha \left(\frac{\sqrt{m/2}}{w} \right) \\ &\geq \frac{1}{16 \cdot 3^\alpha c^2}. \end{aligned}$$

Choose $c = c_\alpha$ so that the last expression is greater than 1. The definition of the Luxemburg norm gives $\|f\|_\Phi \geq w$. $\square$

Lemma 3 (A uniform target-size set contains a full block). *Partition $[n_m]$ into the B_m consecutive blocks of size m . If U_m is uniform among the k_m -element subsets of $[n_m]$, then*

$$\mathbb{P}\{U_m \text{ contains at least one complete block}\} \longrightarrow 1.$$

Proof. Write $n = n_m$, $k = k_m$, and $B = B_m$. If X_t is the indicator that the t -th block is contained in U_m , then

$$p_m := \mathbb{E}X_t = \frac{(k)_m}{(n)_m}, \quad \lambda_m := \mathbb{E} \sum_{t=1}^B X_t = Bp_m,$$

where $(r)_m = r(r-1) \cdots (r-m+1)$.

Put $\delta_m = 1/[e(\log n_m)^\rho]$. Since $k_m/n_m = \delta_m(1+o(1))$ and $m^2/k_m \rightarrow 0$, expansion of the product gives

$$\log p_m = m \log \delta_m + o(1).$$

Also

$$\log n_m = (q_m + 1) \log m, \quad \log \log n_m = \log m + \log \log m + O(1).$$

Consequently,

$$\begin{aligned} \log \lambda_m &= q_m \log m - m - m\rho \log \log n_m + o(m) \\ &= (C - \rho)m \log m - \rho m \log \log m + O(m) \longrightarrow +\infty, \end{aligned}$$

because $C > \rho$. Thus $\lambda_m \rightarrow \infty$.

For two distinct blocks, sampling without replacement gives

$$\mathbb{E}(X_s X_t) = \frac{(k)_{2m}}{(n)_{2m}} \leq p_m^2.$$

Indeed, conditional on containing the first block, the probability of containing the second is $(k-m)_m/(n-m)_m$, and each factor satisfies

$$\frac{k-m-j}{n-m-j} \leq \frac{k-j}{n-j} \quad (0 \leq j < m).$$

For $X = \sum_t X_t$, it follows that

$$\mathbb{E}X^2 \leq \lambda_m + \lambda_m^2.$$

Cauchy-Schwarz applied to $X \mathbf{1}_{\{X>0\}}$ yields

$$\mathbb{P}(X > 0) \geq \frac{(\mathbb{E}X)^2}{\mathbb{E}X^2} \geq \frac{\lambda_m}{1 + \lambda_m} \longrightarrow 1.$$

$\square$

Proof of Theorem 1. Condition on $|J_m| = s$. Exchangeability makes J_m uniform among the s -element subsets of $[n_m]$. For $s \geq k_m$, a uniform s -element set can be coupled to contain a uniform k_m -element set: first sample the s -element set and then choose k_m of its elements uniformly. Hence the probability of containing a complete block is at least the probability in Lemma 3, uniformly for every $s \geq k_m$.

On that event, Lemma 2 gives

$$\mathcal{N}_\Phi(J_m) \geq c_\alpha (\log m)^{\alpha/2}.$$

Finally, $\log m \sim \log \log n_m$. Absorb the asymptotic comparison into c_α and apply Lemma 3. $\square$

4 Verification and novelty notes

The proof uses no computation and no unproved external theorem. The second-moment inequality is proved in line, as are the negative-correlation comparison and the Luxemburg lower bound. The main quantifier to check is that the exchangeable law may vary arbitrarily with m ; conditioning on its cardinality removes that dependence.

A bounded novelty check on August 11, 2026 searched the run indexes for arXiv:2505.00155 and the core endpoint-Orlicz terms, then searched arXiv for the exact title and for combinations of “exchangeable selection”, “uniform fixed-cardinality”, “sampling without replacement”, Orlicz, and bounded orthogonal systems. The search found the source paper and background papers on $\Lambda(\Phi)$ sets and Orlicz sampling, but no later resolution of the source question and no matching exchangeable-selector theorem.

5 Limitations and human-review recommendation

This packet does not prove that every deterministic subset of size k_m is bad. The full-block event is avoidable by a label-dependent choice, so the source’s unrestricted optimality question remains open. The theorem instead shows that replacing independent Bernoulli sampling by any other permutation-invariant random cardinality rule cannot remove the loss.

The result is recommended for expert review as a likely valid substantial partial theorem. The most useful checks are: (i) the asymptotic lower bound for Bp_m , (ii) the negative-correlation inequality for two disjoint blocks, and (iii) the distinction between exchangeable randomized selection and an unrestricted deterministic subset.

References

- [1] W. Burstein, Λ_p Style Bounds in Orlicz Spaces Close to L^2 , arXiv:2505.00155v2, 2025.